

International Islamic University Chittagong
Department of Computer Science and Engineering

B. Sc. in CSE

Final Exam, Spring-2023

Course Code: PHY 1201

Course Title: Physics-II

Time: 2 hours 30 minutes

Full Marks: 50

The figures in the right-hand margin indicate full marks

Part A

[Answer the questions from the followings]

1. a) Define crystal systems. Write the seven crystal systems with example. CO1 R 3
1. b) Show that in a crystal of cubic structure, the distance between the planes with Miller indices h, k, l is equal to $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$, where a is the lattice parameter. CO1 U 5
OR,

Define coordination number. Explain coordination number for NaCl structure.

1. c) Lead is face-centered cubic with an atomic radius of $r = 1.674$ A.U. Find the spacing of (220) planes. CO2 E 2
2. a) How many types of crystal defects in solid? Explain all of the crystal defects. CO1 U 5
OR,
Distinction between metal, insulator and semiconductor in terms of energy band.
2. b) Show that the packing fraction for an fcc lattice is ~ 0.74 and hence write the meaning of this number. CO2 A 5

- Part B

[Answer the questions from the followings]

3. a) State the postulates of the special theory of relativity. CO1 R 2
3. b) Explain "A moving clock always appears to go slow". CO1 U 5
3. c) The length of a spaceship is measured to be exactly half its actual length. Calculate CO2 E 3
(i) the speed of the spaceship
(ii) the time dilation corresponding to one second on the spaceship.
4. a) Calculate the following properties of ${}^{12}_6\text{C}$ particle: CO2 E 7
i. Nuclear Mass ii. Nuclear Size
iii. Nuclear Density iv. Nuclear Charge
v. Nuclear Mass defect vi. Nuclear Binding Energy
vii. Nuclear Binding Energy per nucleon.
4. b) Calculate the glancing angle on the cube (110) of a rock salt crystal ($a = 2.81$ Å) CO2 E 3
corresponding to second order diffraction maximum for the X-rays of wavelength 0.71 Å.
5. a) State the laws of radioactive disintegration and prove it. CO1 U 7
OR,
Show that, the mean life of a radioactive atom is the reciprocal of the radioactive constant.
5. b) The half life of a radioactive substance is 30 days. Calculate CO2 E 3
(i) The radioactive decay constant
(ii) The mean life
(iii) The time taken for $\frac{3}{4}$ of the original number of atoms to disintegrate.

OR,

Calculate the time required for 10% of a sample of thorium to disintegrate. Assume the half-life of thorium to be 1.4×10^{10} years.

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs) of the Questions	
CO1	Understand the basic knowledge of different areas of physics like electromagnetism, structure of matter and modern physics as well as engineering aspect.
CO2	Apply mathematical knowledge to formulate and solve engineering problems.

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part A

[Answer the questions from the followings]

1. a) Define: i) Crystalline solid ii) Crystal CO1 R 2
 1. b) What is a packing fraction? With proper illustration, calculate packing fraction for a FCC crystal structure. CO1 U 5

OR

Define coordination number. Explain coordination number for NaCl structure.

1. c) The lattice constants of a simple lattice is a . Find the lattice spacings between (111), (112) and (113) lattice planes. CO2 A 3
 2. a) Define Miller indices and draw the plane of (001), (110), (011) and (111). CO1 U 5
 2. b) Show that in a crystal of cubic structure, the distance between the planes with Miller indices h, k, l is equal to $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$, where a is the lattice parameter CO2 A 5

OR,

- 2 b) Distinguish between metal, insulator and semiconductor in terms of energy band theory.

Part B
[Answer the questions from the followings]

- | | | | |
|---|-----|---|---|
| 3. a) What are the postulates of the special theory of relativity? | CO1 | R | 2 |
| 3. b) Deduce Einstein's mass-energy equation, and mention its significance. | CO1 | U | 5 |

OR

Explain the following quotation "A moving clock always appears to go slow".

- | | | | |
|--|-----|---|---|
| 3. b) Calculate the mass of the electron when it is moving with a K.E. of 10MeV. | CO2 | A | 3 |
|--|-----|---|---|

4. a) Calculate the following properties of ${}_{28}\text{Ni}^{64}$ particle:
- i. Nuclear Mass
 - ii. Nuclear Size
 - iii. Nuclear Density
 - iv. Nuclear Mass defect
 - v. Nuclear Binding Energy

	CO2	A	5
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OR,

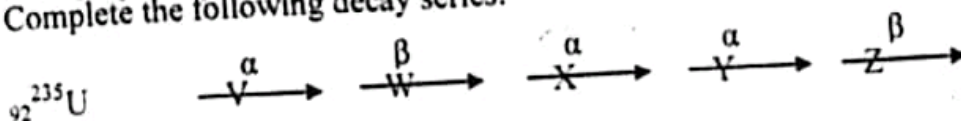
How to calculate the radius on a Helium atom, using Bohr atomic model?

4. b) Draw the nuclear binding energy curve.

	CO1	R	2
--	-----	---	---

4. c) Complete the following decay series:

	CO1	U	3
--	-----	---	---



Write down the values of VWXYZ in the above series.

- | | | | |
|---|-----|---|---|
| 5. a) What do you mean by half-life of a radioactive substance? Hence obtain an expression for the half-life. | CO1 | U | 5 |
|---|-----|---|---|

OR,

State and explain Radio-active decay laws.

5. b) The half life of a radioactive substance is 30 days. Calculate

	CO2	A	5
--	-----	---	---

- (i) The radioactive decay constant
- (ii) The mean life
- (iii) The time taken for $\frac{3}{4}$ of the original number of atoms to disintegrate
- (iv) The time for $\frac{1}{8}$ of the original number of atoms to remain unchanged.

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(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Part A

[Answer the questions from the followings]

- 1 a) Distinction between metal, insulator and semiconductor in terms of energy band diagram. CO1 U 5

Or,

- 1 a) Define: i) Lattice ii) Basis iii) Crystal iv) Unit Cell v) Crystalline Solid CO1 R 5
1. b) Show that the packing fraction for a fcc lattice is ~0.74 and hence write the meaning of this number. CO2 A 5

2. a) Define Miller indices and draw the plane of (100), (010), (101) and (111). CO1 U 5
2. b) Show that in a crystal of cubic structure, the distance between the planes with Miller indices h, k, l is equal to $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$, where a is the lattice parameter. CO2 A 5

Or,

- 2 b) If a plane of a cubic lattice intersects the X, Y and Z axes at 3, 2 and 2 respectively. If lattice parameter (X+1) and find the distance of the plane from origin and draw this plane. Here X is the last digit of your student ID. CO2 E 5

Part B

[Answer the questions from the followings]

- | | | | | | |
|-----|----|---|-----|---|---|
| 3. | a) | Explain "A moving clock always appears to go slow". | CO2 | A | 6 |
| 3. | b) | The length of a spaceship is measured to be exactly half its actual length. Calculate | CO2 | A | 4 |
| | | (i) the speed of the spaceship | | | |
| | | (ii) the time dilation corresponding to one second on the spaceship. | | | |
| 4. | a) | State the laws of radioactive disintegration. Show that, the number of radioactive atoms decreases exponentially with time. | CO2 | A | 5 |
| 4. | b) | The half-life of a radioactive substance is 30 days. Calculate (i) the radioactive decay constant, (ii) the mean life and (iii) the time taken for (1/8) of the original number of atoms to remain unchanged. | CO2 | E | 5 |
| 5. | a) | Define 'Nuclear Fission' and 'Nuclear Fusion' with an example of each. | CO1 | A | 4 |
| 5. | b) | Find the nuclear Mass, Size, density, binding energy and binding energy per nucleon for ${}_{88}\text{Ra}^{226}$. | CO2 | A | 6 |
| Or, | | | | | |
| 5. | a) | Write short note on:
(i) Photoelectric effect
(ii) Compton effect | CO1 | R | 4 |
| 5. | b) | Show that, the mean life of a radioactive atom is the reciprocal of the radioactive constant. | CO2 | R | 6 |

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn- 2018(Special)
Course Code: PHY- 1201 Course Title: Physics-II
Total marks: 50 Time: 2 and half hours

[Answer any *two* questions from **Group-A** and any *three* questions from **Group-B**;
 Separate answer script must be used for Group-A and Group-B.]

Group-A

- | | | |
|-------|--|---|
| 1. a) | What is packing fraction? | 1 |
| b) | Find out the packing fraction of | 6 |
| | (i) Simple cubic Crystal | |
| | (ii) Body centered cubic crystal. | |
| c) | The lattice constants of a simple lattice is a . Find the lattice spacing between (101), (211) and (113) lattice planes. | 3 |
| 2.a) | What is crystal defects? | 1 |
| b) | Explain different types of Point defects in crystal with proper diagram. | 5 |
| c) | Write short notes on i) Metallic bond and ii) Covalent bond. | 4 |
| 3. a) | What is co-ordination number? | 1 |
| b) | Explain co-ordination number for NaCl structure. | 3 |
| c) | Draw the planes (110), (101) and (111). | 3 |
| d) | Define: (i) Crystal (ii) Lattice (iii) Basis | 3 |

Group-B

- | | | |
|-------|---|---|
| 4. a) | State the postulates of the special theory of relativity. | 2 |
| b) | Explain "A moving clock always appears to go slow". | 5 |
| c) | The mean life of π meson is 2×10^{-8} Sec. Calculate the mean life of a meson moving with a velocity of .8 c. | 3 |
| 5. a) | What is an X-rays? Write down the properties of X-rays. | 2 |
| b) | Derive Bragg's Law for X-ray diffraction in crystals. | 5 |
| c) | If X-rays of wavelength $0.5\text{\AA}$ are detected at an angle of 5° in the first order, what is the spacing between the adjacent planes of the crystal? | 3 |
| 6. a) | State the laws of radioactive disintegration. | 2 |
| b) | What is half-life period? Derive an expression for the half -life period. | 5 |
| c) | 1 gram of radium is reduced by 2.1 mg in 5 years by α -decay. Calculate the half-life period of radium. | 3 |
| 7. a) | Define nuclear binding energy. | 1 |
| b) | Calculate the binding energy for deuteron. | 3 |
| c) | Define 'Nuclear Fission' and 'Nuclear Fusion' with an example of each. | 4 |
| d) | Draw nuclear binding energy curve. | 2 |

[Answer any *two* questions from **Group-A** and any *three* question from **Group-B**;
 Separate answer script must be used for Group-A and Group-B;
 Figures in the right hand margin indicate full marks.]

Group-A

- | | | |
|-------|---|---|
| 1. a) | Define the terms: (i) Lattice (ii) Basis (iii) Crystal Structure | 3 |
| b) | What is packing fraction? Find out the packing fraction of face centered cubic crystal. | 5 |
| c) | Lead is face-centered cubic with an atomic radius of $a = 1.564$ A.U. Find the spacing of (220) planes. | 2 |
| 2. a) | What is crystal defect. | 1 |
| b) | Explain different types of defects in crystal.. | 6 |
| c) | Write down shorts on: i) Covalent bond and ii) Metallic bond. | 3 |
| 3. a) | Define single crystal and poly crystal with diagram. | 2 |
| b) | Draw the planes (001), (110) and (111) | 3 |
| c) | Distinction between metal, insulator and semiconductor in terms of energy band. | 5 |

Group-B

- | | | |
|-------|---|---|
| 4. a) | State the postulates of the special theory of relativity. | 2 |
| b) | Explain "A moving clock always appears to go slow". | 5 |
| c) | The time of an unstable particle at rest is 10^{-6} s. If at the instant of creation, it moves with a speed of $0.9c$, then measure its relativistic time. | 3 |
| 5. a) | What is an X-rays? Write down the properties of X-rays. | 2 |
| b) | Derive Bragg's Law for X-ray diffraction in crystals. | 5 |
| c) | The distance d between two adjacent atomic planes in a calcite is 3.04 A.U. Find the wavelength for which the first order reflection occurs at 15.7° . | 3 |
| 6. a) | State the laws of radioactive disintegration. | 2 |
| b) | What is mean life of radioactive element? Derive an expression for mean life. | 5 |
| c) | 1 gram of radium is reduced by 0.1 mg in 10 years by α -decay. Calculate the half-life period of radium. | 3 |
| 7. a) | What is binding energy? | 1 |
| b) | Calculate the binding energy of an α -particle. | 2 |
| c) | Calculate Nuclear Density for ${}_{88}\text{Ra}^{226}$. | 3 |
| d) | Define 'Nuclear Fission' and 'Nuclear Fusion' with an example of each. | 4 |